DESIGN AND NUMERICAL ANALYSIS OF A SUB-SURFACE TWO-BODY WAVE ENERGY CONVERTOR

Virag Mishra, Michael Kelly, Joseph Piacenza, and Solomon C. Yim

***Abstract*—Recent developments in self-reacting oscillating Wave Energy Convertors (WECs) point toward two popular design concepts. These designs are either semi-submerged system in which a surface piercing float reacts against a relatively still heave plate or a single body system that reacts against the ocean floor. The first design employs a rigid spar connection between the actuator and the reaction body; however, the second design uses a flexible belt connection between the actuator and the foundation. In this work, a two-body sub-surface WEC is proposed in which the submerged float reacts against a base that is dynamically active. The dynamics of the reacting body is influenced by both the buoyancy parameters and the PTO generated tether tension. As a result, the two-body system oscillates in an antipodal mode. The optimal hydrostatic conditions that result in a design that can generate mechanical power, in single degree of freedom motion, under the wave excitation are discussed in this work. The hydrodynamic analysis is carried out for different wave conditions observed off the coast of US. Optimal PTO parameters are determined for various designs. The analysis is carried out using Ansys-Aqwa. Observations of the relative predictive performance between the two numerical simulations models are delineated.**

***Index Terms*—Hydrodynamics, Power Take Off (PTO), Wave Energy Converter (WEC)**

I. INTRODUCTION

DUE to the concerted effort to achieve energy independence as well as to reduce the greenhouse gas emission, most of the development to date in offshore renewable energy systems has been targeted at "utility-scale" or "community-scale" devices. These devices have electricity generation capacities in the order of MW and kW respectively. However, another application area that is attracting investment in wave energy converters is "small-scale" devices. These devices can provide autonomous power to various ocean-bound systems, such as autonomous underwater vehicles [1], observation systems [2], and even aquaculture farms. The expectation is that these devices can be used to power the on-site stand-alone systems which are located remotely, resulting in exponential growth in the sector often referred as "Blue economy" [3], [4]. This paper provides an overview of the initial design process of such a device. First, the primary dimension of the device is derived based on the constraints of the operating design (Sect. yyy). The details of the deployment sites are discussed in (Sect. xxxxxx). The results and the following discussion are presented in (Sect. zzzz). This project is funded by the NAVSEA, as a one-year grant to design a sub-surface WEC capable of continuous power generation of 100W at various sites with distinct wave characteristics off the coast of the United States.

The authors would like to thank Department of Defense, NAVSEA-BEST, NSWCCD, NIWC PAC, NAVFAC EXWC and U.S. DoD contractor Battelle. This work is supported under NAVSEA Contract Number: N66001-18-9-4703.

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II. BACKGROUND

Numerical benchmarking of eight WECs with different working principles was conducted to evaluate their power matrices and site specific annual energy production [5]. Out of these, two operated close to seabed and only one was near surface submerged WECs. This shows that the focus has mainly been around floating devices. Some of the popular floating WECs with heave plate are MBARI-WEC [6], OPT Powerbuoy, OSU-Lupa (Laboratory Upgrade Point Absorber) [7] and Wavebob [8].

Though the wave energy decays exponentially with depth in deep water regime, subsurface WECs provide the potential benefit of survivability and covertness. Possibly these are reasons why subsurface WECs are emerging as a preferred design choice for WEC developers. Some of the popular designs that are either bottom fixed or do not have a heave plate as reacting bodies are AWS Ocean Energy's Archimedes Waveswing, Symphony Wave Power's WEC, Carnegie Clean Energy's CETO 6 [9], and CalWave's xWave. CETO6 and xWave are multi tethered, multi DOF point absorbers which operate close to the surface. The Waveswing is an air pressure differential point absorber tethered to the sea bed and Symphony Wave Power's WEC is a bottom-fixed device operating in shallow waters. Conceptual submerged WEC capable of producing 10 W of power with a float and six trubines has been proposed in [10].

In all the buoyant designs mentioned above the buoyancy of the float acts as the actuation force and either the foundation in the sea bed or the heave plate acts as a reaction body. In this work we propose a two-body submerged system with a float and heave plate connected by a tether and the entire system is moored to the foundation. The float will act as an actuator and

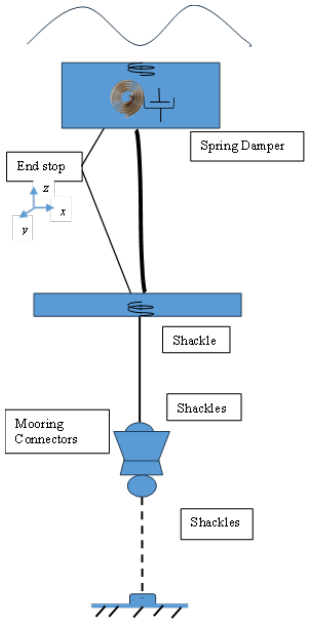


Fig. 1. Sketch of the two-body sub-surface WEC under analysis

the buoyancy of the heave plate will aid to power generation. This work will evaluate the performance for the above described WEC and compare six different design candidates at different wave energy sites in the US.

III. Device Details

The WEC design under study is a two-body point absorber consisting of a submerged float and a heave plate. The design of the WEC supports the sub-surface operation.

IV. Wave Climate at deployment locations

Based on the works of [11] on wave resource assessment off the coast of US North West, a set of guidelines were developed and included in a technical specification for wave resource assessment by International Electrotechnical Commission (IEC) [12]. The IEC brings together international marine energy experts to develop consensus-based standards that guide the development of a broad range of sectors. The current industry consensus is determination of representative sea states based on the energy weighted joint probability distribution function of primarily H_s and T_e/T_P [13]. However, the true description of a sea state should involve additional parameters such as spectral shape, peak direction, directional spreading, wind, tide, and current; yet data to fully characterize these aspects are often unavailable. Even when such data exist, current approaches neither develop a multidimensional probabilistic description of the deployment site nor account for energy-weighted probabilities, and they typically preclude the evaluation of WEC performance across multiple deployment climates. A need for developing a methodology that could account for multiple deployment locations was presented in a Wave Energy Prize competition by US Department of Energy in 2016. A new methodology was proposed in this competition

TABLE I
NDBC BUOYS CHOSEN FOR ANALYSIS

NDBC Buoy	US Location	Depth [m]	Year
51207	Oahu, HI	81	2019
42001	Mid Gulf, New Orleans, LA	32	2018
46050	PACWAVE, OR	149	2017
46254	San Diego, CA	46	2019
41009	Canaveral, FL	42	2011
44018	Cape Cod, MA	44	2010

by [14] that applied a cluster analysis to scatter diagrams in order to determine a subset of sea states that could be scaled to find the average annual power flux at each wave climate. A common centroid representing of H_s and T_P was determined for the clusters. This procedure opens the possibility of using a single set of cluster centroid for different locations. Since WEC performance assessment simulations are computationally resource intensive, in this work the methodology described above is used to run simulations for a single set of cluster centroids and the power produced for each centroid is scaled depending on the site under consideration.

A set of 18 sea states were considered in this report for all the analysis. The spectrum used to generate the wave is Bretschneider Spectrum. Although these different wave data are not all from the same physical conditions (e.g. depths range from 42 m to 3200 m) it provides an idea of the WEC performance in varied wave profiles, using the same data for each. Note that if using the weights in the 18 sea states, it makes up varying percentages of the total joint probability diagram ($67\% - 95\%$). This is because in some locations there can be significant portions of the year in sea states not represented by the 18 sea states.

Sea states that are representative of the U.S. Coast wave climates were selected. The locations were chosen to determine site specific optimal PTO coefficient for the WEC. In addition, these sites either host a testing facility or a deployment berth and are probable sites for the development of WEC farms. Sea state parameters reported by NDBC were used to analyze the wave climate at each buoy. The significant wave height reported by NDBC (and used here) is the average of the highest one-third of all the wave heights during a 20-minute sampling period. The peak period is the period with maximum energy in the spectrum. Lastly, the peak direction was used to determine relative wave scatter diagrams for each of the deployment locations. The list of the buoy locations chosen for the analysis is presented in Table I.

A cluster analysis was applied to wave scatter diagrams to identify a representative subset of sea states that can be scaled to estimate the average annual power (AAP) flux. The subset is determined by minimizing the distance between each data point and its corresponding cluster centroid. Four pdist2 options for minimizing the distance function were compared these are Squared Euclidean, Standardized Euclidean, Eclidean and Mahalanobis. This clustering is performed using

the k-means algorithm implemented in the MATLAB Statistics Toolbox, where the number of clusters and the centroid values is specified by the user. The algorithm employs an iterative partitioning procedure that optimizes the associated memberships by minimizing the sum of point-to-centroid distances across all k clusters. As a result, each sea state is assigned to the cluster whose distance to centroid yields the minimum value, providing an optimally representative classification of the wave climate. In this paper results corresponding to Mahalanobis method are presented.

V. Some common mistakes

The word 'data' is plural, not singular. A graph within a graph is an 'inset,' not an 'insert.' The word 'alternatively' is preferred to the word 'alternately' (unless you really mean something that alternates). Use the word 'whereas' instead of 'while' (unless you are referring to simultaneous events). Do not use the word 'essentially' to mean 'approximately' or 'effectively.' Do not use the word 'issue' as a euphemism for 'problem.' Be aware of the different meanings of the homophones 'affect' (usually a verb) and 'effect' (usually a noun), 'complement' and 'compliment,' 'discreet' and 'discrete,' 'principal' (e.g., 'principal investigator') and 'principle' (e.g., 'principle of measurement'). Do not confuse 'imply' and 'infer.' Prefixes such as 'non,' 'sub,' 'micro,' 'multi,' and 'ultra' are not independent words; they should be joined to the words they modify, usually without a hyphen. There is no period after the 'et' in the Latin abbreviation 'et al.' (it is also not italicized). The abbreviation 'i.e.,' means 'that is,' and the abbreviation 'e.g.,' means 'for example' (these abbreviations are not italicized). The suffix '-ize' is correct, both phonetically and etymologically, in both British and American English and there is no reason to use '-ise.' For further information, please see the relevant entry in the Oxford English Dictionary.

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A. Types of graphics

The following list outlines the different types of graphics published in IMEJ journals. They are categorized based on their construction, and use of colour or shades of grey:

1) Colour/grey-scale figures: Figures that are meant to appear in colour, or shades of black/grey. Such figures may include photographs, illustrations, multicolour graphs, and flowcharts.

2) Line art figures: Figures that are composed of only black lines and shapes. These figures should have no shades or half-tones of grey, only black and white.

3) Tables: Data charts which are typically black and white, but sometimes include colour.

4) Multipart figures: Figures compiled of more than one sub-figure presented side-by-side, or stacked. If a multipart figure is made up of multiple figure types (one part is line-art, and another is grey-scale or colour) the figure should meet the stricter guidelines.

B. File formats for graphics

Format and save your graphics using a suitable graphics processing program that will allow you to export images as PostScript (PS), Encapsulated PostScript (.EPS), Tagged Image File Format (.TIFF), Portable Network Graphics (.PNG) or JPEG (.JPG) at suitable resolution (see below). It is strongly recommended that graphics be saved in PNG format rather than DOC, XLS, or PPT native formats. Doing so will protect your figures from common problems that occur when working on the files across multiple platforms. When submitting your final paper, your graphics should all be submitted individually in one of these formats along with the manuscript.

C. Sizing of graphics

Most charts, graphs, and tables are one column wide (86 mm) or page wide (178 mm). The maximum depth a graphic can be is 250 mm. When choosing the depth of a graphic, please allow space for a caption. Figures can be sized between column and page widths if the author chooses, however it is recommended that figures are not sized less than column width unless when necessary. IMEJ uses page size A4 with a column measurement of 86 mm. The column spacing is 6 mm. The left and right margins are 16 mm, top margin 22 mm (including header) and bottom margin 14 mm. The header height is 10 mm.

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The proper resolution of your figures will depend on the type of figure it is as defined in the 'Types of Figures' section. Photographs, and colour and grey-scale figures should be at least 300 dpi. Line art, including tables should be a minimum of 600 dpi. The JPEG format should only be used for photographs; PNG format should be used for diagrams as it preserves sharp boundaries.

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In order to preserve the integrity of figures across multiple computer platforms, we accept vector graphics in the following formats: .EPS/.PDF/.PS. All fonts must be embedded or text converted to outlines in order to achieve the best-quality results.

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The term colour space refers to the entire sum of colours that can be represented within the said medium. For our purposes, the three main colour spaces are: grey-scale, RGB (red/green/blue) and CMYK (cyan/magenta/yellow/black). RGB is generally used with on-screen graphics, whereas CMYK is used for printing purposes. Colour figures should be generated all in RGB or all in CMYK colour space. grey-scale images should be submitted in grey-scale colour space. Line art may be provided in grey-scale OR bitmap colour space. Note that 'bitmap colour space' and 'bitmap file format' are not the same thing. When bitmap colour space is selected, .PNG or .TIFF are the recommended file formats.

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J. Referencing a figure or table within your paper

When referencing your figures and tables within your paper, use the abbreviation 'Fig.' even at the beginning of a sentence. Do not abbreviate 'Table.' Tables should be numbered with Roman numerals.

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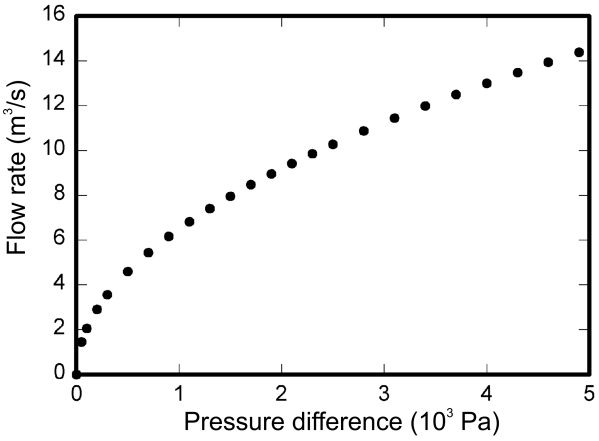


Fig. 2. Flow rate as a function of pressure difference. Note that "Fig." is abbreviated. There is a period after the figure number, followed by two spaces. It is good practice to explain the significance of the figure in the caption.

TABLE II
EXAMPLE VARIABLES AND UNITS FOR MARINE RENEWABLE ENERGY

Symbol	Quantity	Unit
U	Flow speed	1 m/s = $0.51\dot{4}$ kn[a]
p	Pressure	Pa
μ	Dynamic viscosity	Pa·s
ν	Kinematic viscosity	m^2/s
L	Distance	1000 m = 1 km = $\frac{250}{463}$ nmi[a]
E	Energy	1 kJ = $0.27\dot{7}$ kWh
a	Acceleration	m·s^{-2}
F	Force	N
P	Power	W
M	Mass	1000 kg = 1 t
Q	Flow rate	m^3/s

Vertical lines in tables are deprecated. Statements that serve as captions for the entire table do not need footnote letters.
[a] The Admiralty measured mile and Admiralty knot should not be used to avoid confusion.

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As the IMEJ is an online-only journal, an author can publish colour figures on IMEJ at no charge. However, authors may wish to bear in mind that readers will often print journal articles in monochrome.

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A. References

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the last name [15]. Use them all; use et al. only if names are not given [16]. Abbreviate conference titles. When referencing a patent, provide the day and the month of issue, or application. References should include enough information for the reader to find the original source. Do not combine references. There must be only one reference with each number [17]. If there is a URL included with the print reference, it can be included at the end of the reference [5], [17].

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X. Publication principles

The IMEJ publishes peer-reviewed articles covering fundamental and applied research, including: case studies; field deployment and testing; technical, economic and social analysis; and laboratory based studies covering wave, tidal and ocean thermal energy conversion. Authors should consider the following points:

1) Technical papers submitted for publication must advance the state of knowledge and must cite relevant prior work.
2) The length of a submitted paper should be commensurate with the importance, or appropriate to the complexity, of the work. For example, an obvious extension of previously published work might not be appropriate for publication or might be adequately treated in just a few pages.
3) Authors must convince both peer reviewers and the editors of the scientific and technical merit of a paper; the standards of proof are higher when extraordinary or unexpected results are reported.
4) Because replication is required for scientific progress, papers submitted for publication must provide sufficient information to allow readers to perform similar experiments or calculations and use the reported results. Although not everything need be disclosed, a paper must contain new, useable, and fully described information. Authors should expect to be challenged by reviewers if the results are not supported by adequate data and critical details.
5) Papers that describe ongoing work or announce the latest technical achievement, may not be appropriate for publication. However, they may suitable for presentation at the European Wave and Tidal Energy Conference (ewtec.org), for example.

XI. Discussion and Conclusion

This study investigated the hydrodynamic performance and design optimization of a submerged wave energy converter (WEC) equipped with a tether–PTO system. The report first introduced the overall system architecture, including the prime movers, tether specifications, PTO algorithm, and PTO parameters. A numerical modelling framework was then developed, covering the governing theory, model implementation, verification of the developed UDF and the selection of representative sea states for performance assessment.

The mechanical average annual power capture at six different sites along the US coasts for different designs is compared.

A comprehensive sensitivity analysis was conducted under both survival and operational sea states. The operational-state analysis examined the influence of key design parameters for six different designs of float and heave plate. These designs vary in characteristic dimensions. Several sensitivity analyses were carried out for different operating parameters, primarily the PTO coefficients, draft, initial tether length, mass ratio between float and heave plate and mass of heave plate. The results demonstrate that the hydrodynamic response and power capture of the submerged WEC are strongly influenced by both the system configuration and the PTO parameters. The sensitivity analyses showed that careful selection of key design parameters, including the geometric configuration, mass distribution, and the initial float–heave plate gap, can substantially enhance energy capture while maintaining acceptable structural loads and system stability. For a configuration that is submerged 2 m below the surface, the power capture at most of the sites for most of the PTO coefficients is more than the minimum targeted value (only location where power was lower was in San Diego). The variation of power capture at various depths and different tether lengths is also compared for two designs. A linear drop in the captured power is observed with an increase in the draft and increase in the initial tether length (over the range of 2-7 m submerged). The optimal PTO coefficient at six different sites around US were found out for maximum power capture.

Overall, the validated numerical model and the comprehensive parametric analyses presented in this study provide a robust framework for optimizing submerged WEC designs, enabling an effective balance between energy production, structural integrity, operational reliability and survivability for site-specific deployment.

Appendix A
The Lanchester-Betz limit in a relativistic rotating reference frame

Appendices, if needed, appear before the acknowledgement.

Appendix B

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Acknowledgement

The authors would like to thank Department of Defense, NAVSEA-BEST, NSWCCD, NIWC PAC, NAVFAC EXWC and U.S. DoD contractor Battelle. This work is supported under NAVSEA Contract Number: N66001-18-9-4703.
Disclaimer: Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the Department of Defense.

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